

Modeling Toronto Apartment Rental Prices

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Introduction

This section should describe the background and objectives of the study.

Data Description

This section should describe where the dataset comes from, the sample size, how the data were collected, etc.

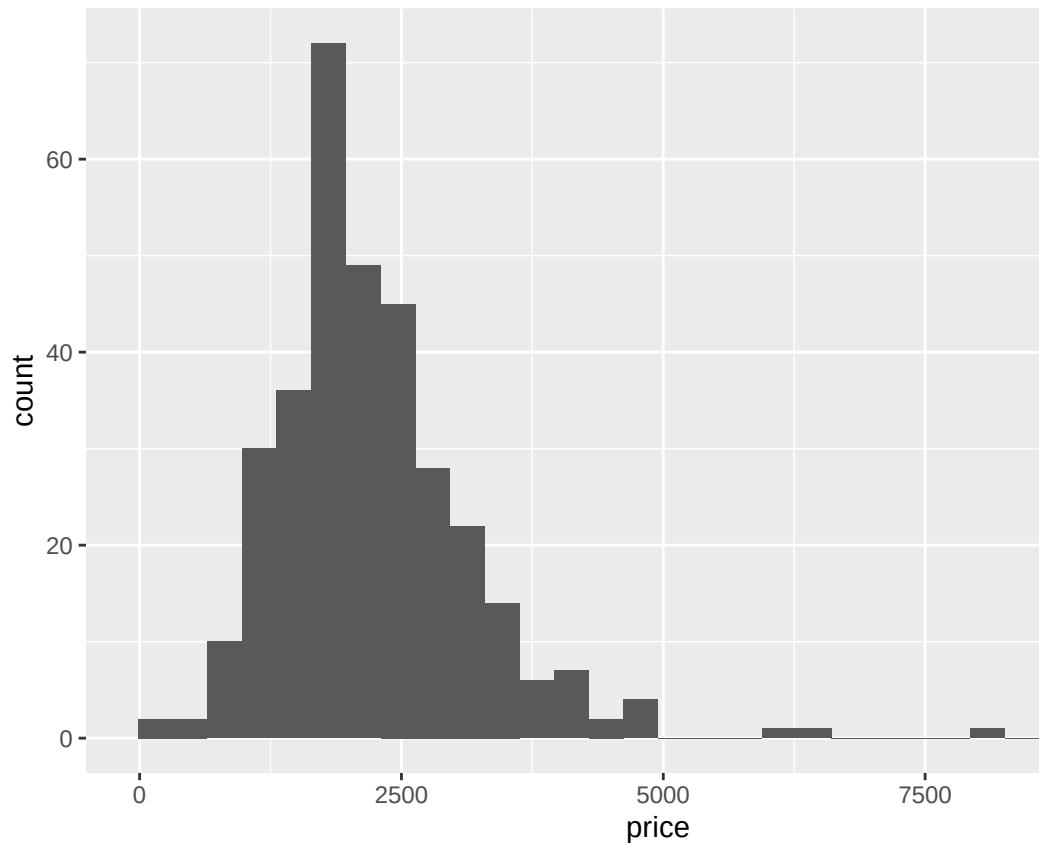
Our data was obtained from <https://www.kaggle.com/datasets/rajacsp/toronto-apartment-price>. The total number of rows in the original dataset is 1,124.

Exploratory Data Analysis (EDA)

The total number of rows in the original dataset is 1,124. During data cleaning, we removed 308 duplicates. We also removed 5 rows with unusual values for `price` (\$65, \$99, \$99, \$36,900, \$535,000). The sample size is now `n = 811`. There are no missing values.

Before modeling, we want to find out more about the data. The variables we're most interested in are `price`, `bedroom`, `bathroom`, and `den`.

Below are the summary statistics for `price` in a table.

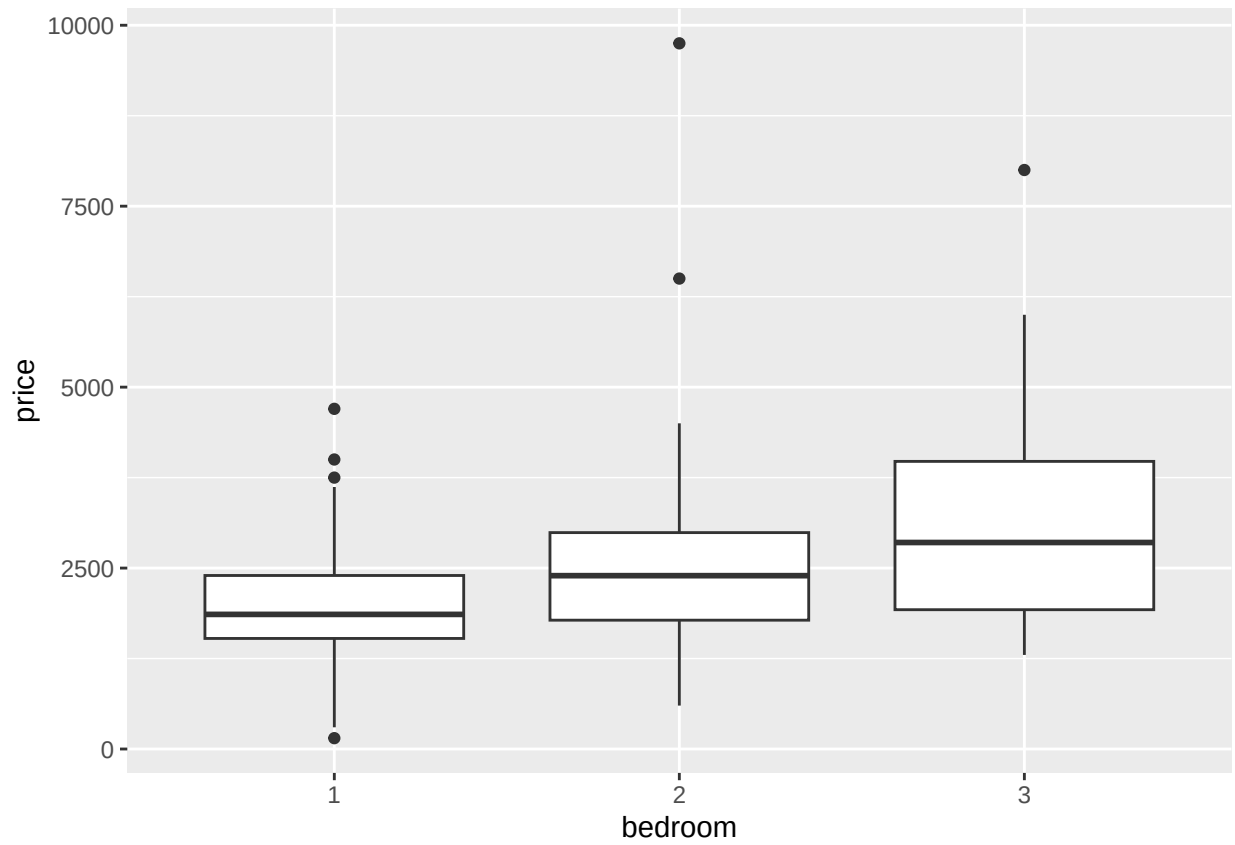


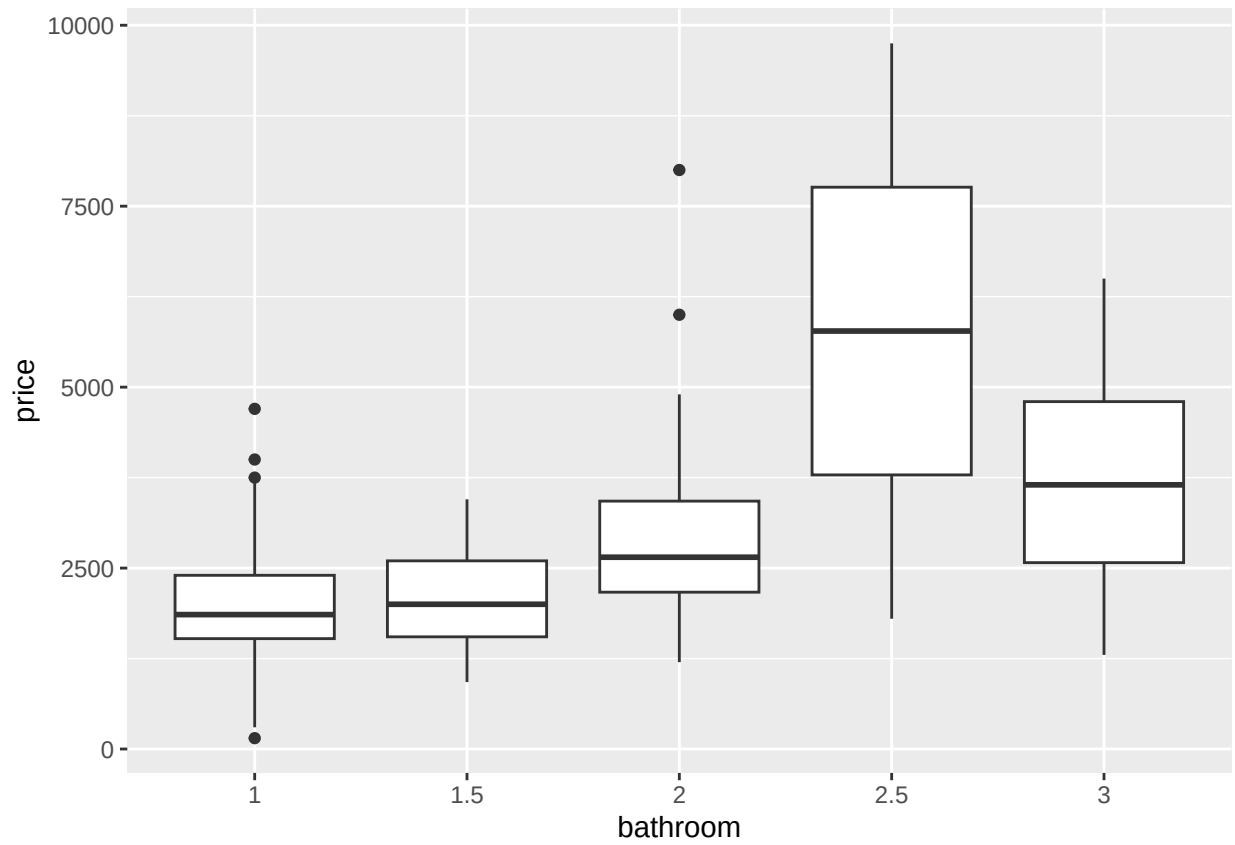
This histogram of price shows...

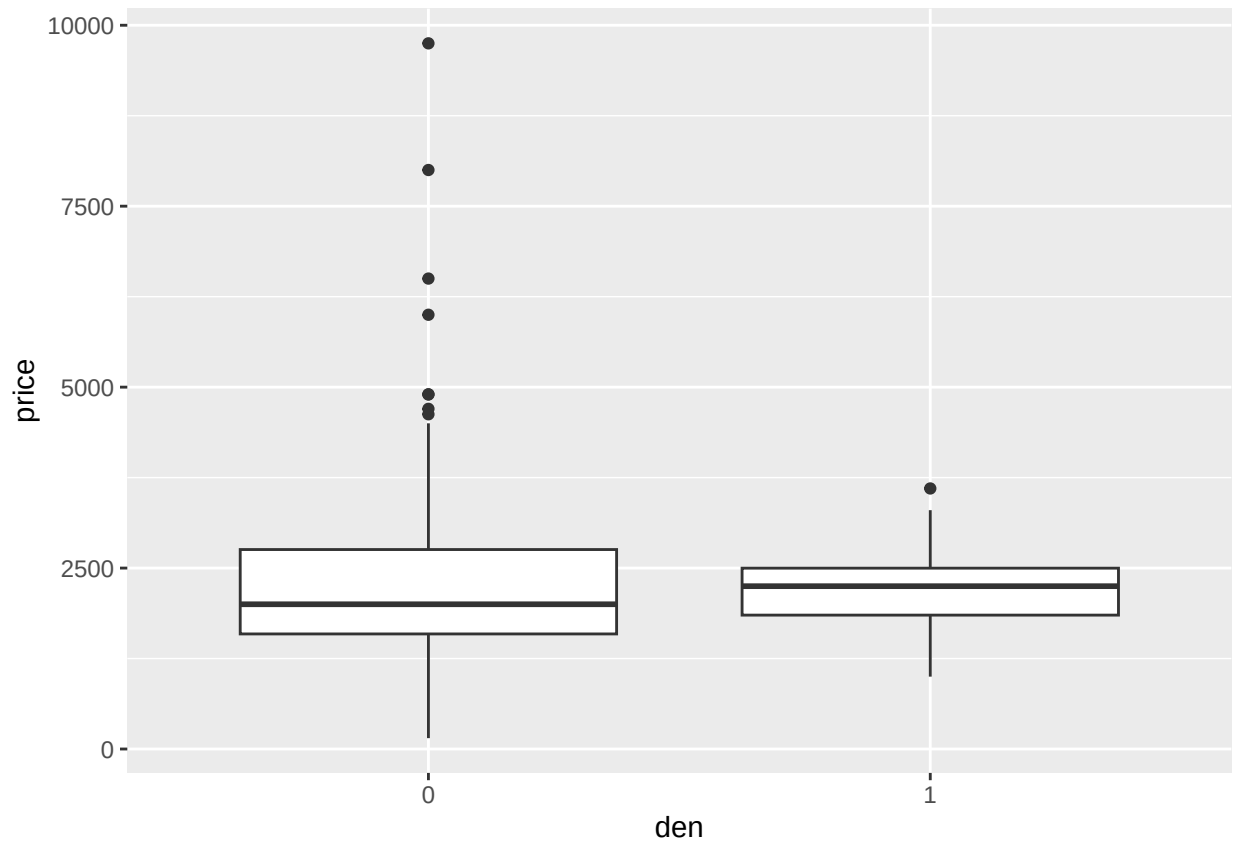
```
##
##  1  2  3
## 186 113 34
```

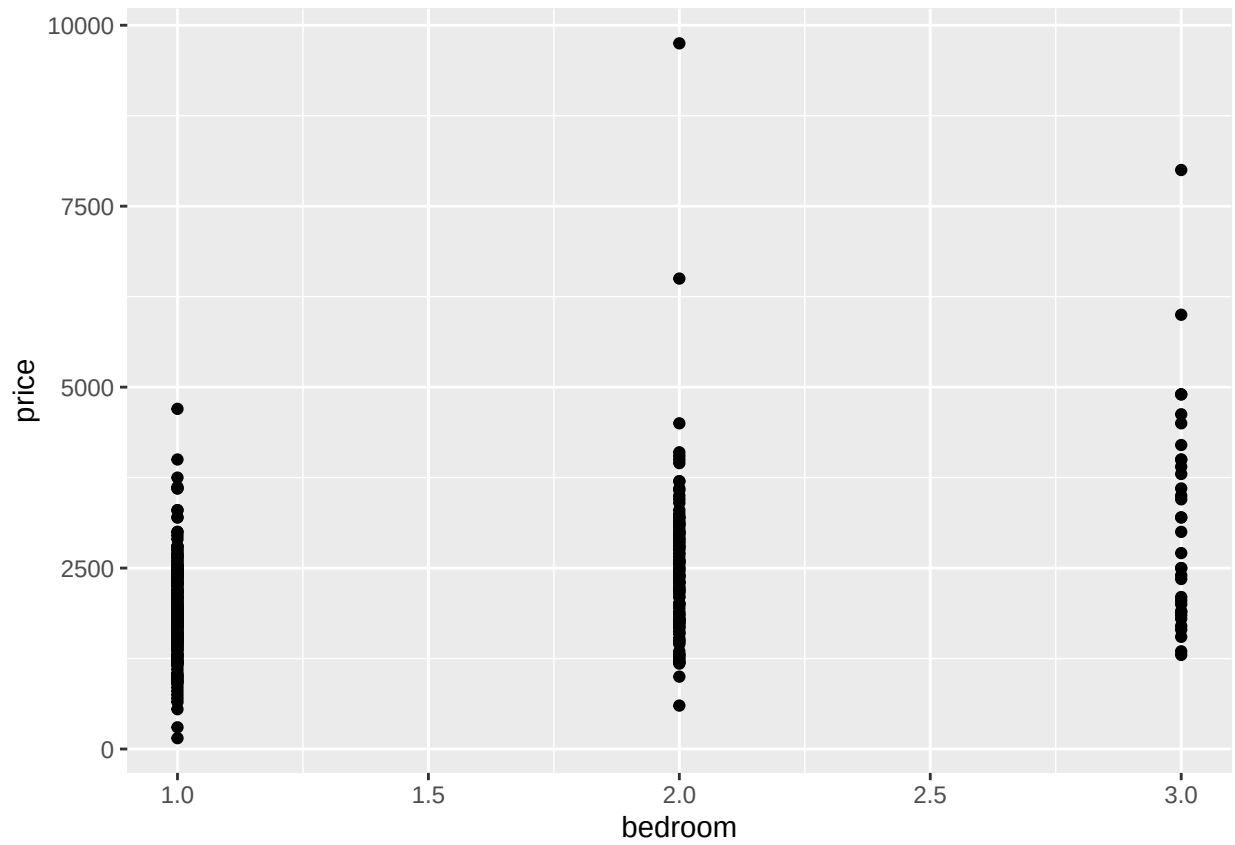
```
##
##  1 1.5  2 2.5  3
## 221  21  83  2   6
```

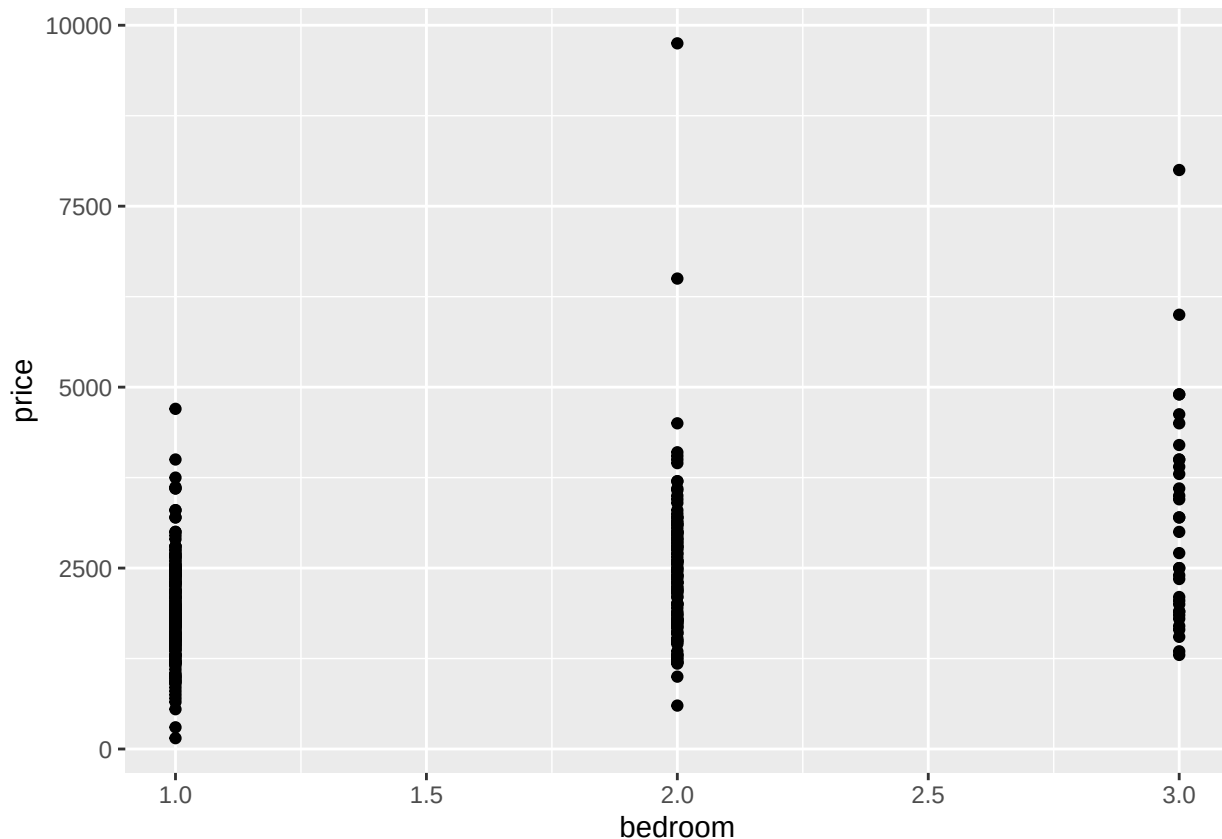
```
##
##  0  1
## 276 57
```











```
##          bedroom  bathroom      den      price
## bedroom  1.0000000  0.50147207 -0.36728231  0.37884260
## bathroom 0.5014721  1.00000000 -0.09791753  0.46537487
## den      -0.3672823 -0.09791753  1.00000000 -0.01986721
## price     0.3788426  0.46537487 -0.01986721  1.00000000
```

Models

This section should include at least two statistical models for the same dataset, justifications of the choices of the models, model diagnostics, parameter estimates and their standard errors and p-values, and interpretation of the results.

```
##
## Call:
## lm(formula = price ~ ., data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2661.7  -517.0   -45.2    441.3   6515.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    677.50     160.03   4.233 2.99e-05 ***
## bedroom         360.51      89.79   4.015 7.36e-05 ***
## bathroom        734.23     114.15   6.432 4.43e-10 ***
```

```
## den          277.02      139.37    1.988    0.0477 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 885.9 on 329 degrees of freedom
## Multiple R-squared:  0.2538, Adjusted R-squared:  0.247
## F-statistic: 37.3 on 3 and 329 DF,  p-value: < 2.2e-16
```

Conclusion

This section should provide overall conclusions, any limitations in the datasets and analyses, and validity of the conclusions.

References

Any references can go here.

Appendix

```
library(tidyverse)
```